LECTURE 14 — Quality Management & Continuous Improvement

Fit for purpose

Meets requirements

Customer satisfaction

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Quality vs Grade

Quality: Meets exact requirements

Zero defects, functions perfectly.

Grade: Level of features

Basic model vs. Luxury model.

Low Grade + High Quality = PMI Acceptable.

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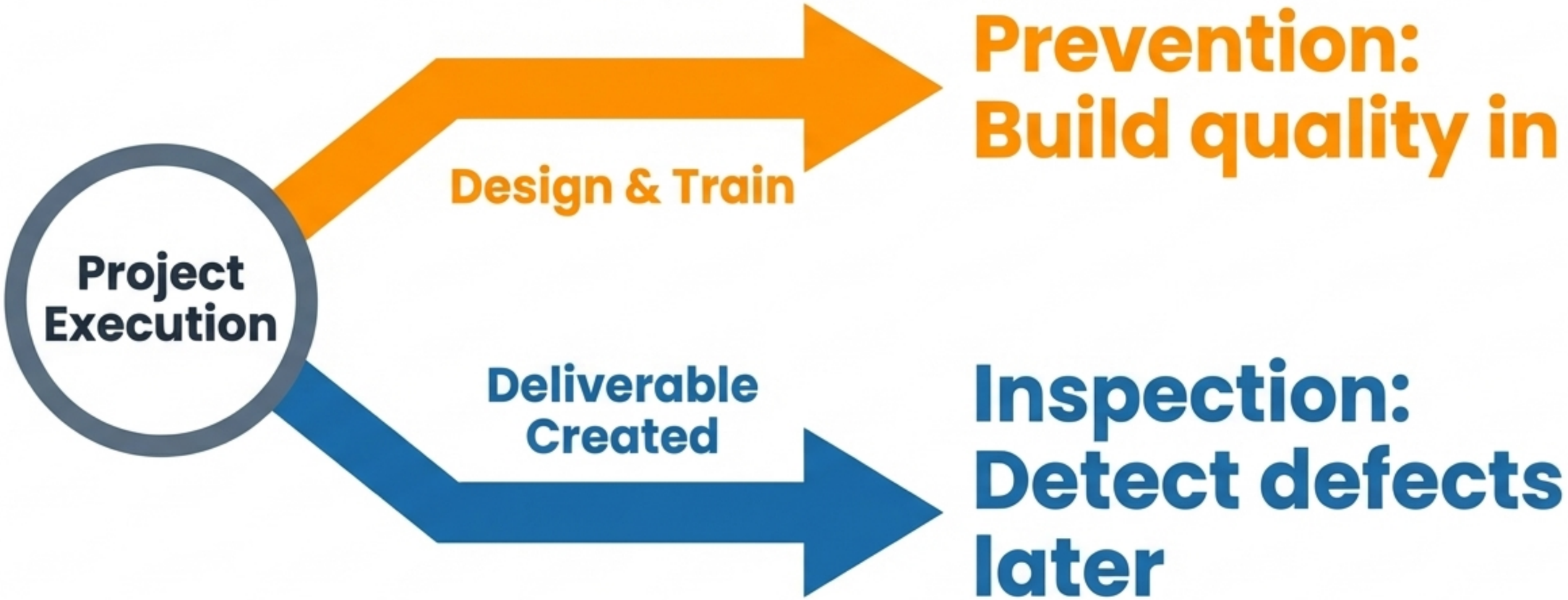
Prevention vs Inspection



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Cost of Quality

Prevention Cost

Training, Equipment

Appraisal Cost

Testing, Inspections

Failure Cost

Rework, Scrap, Warranty Claims

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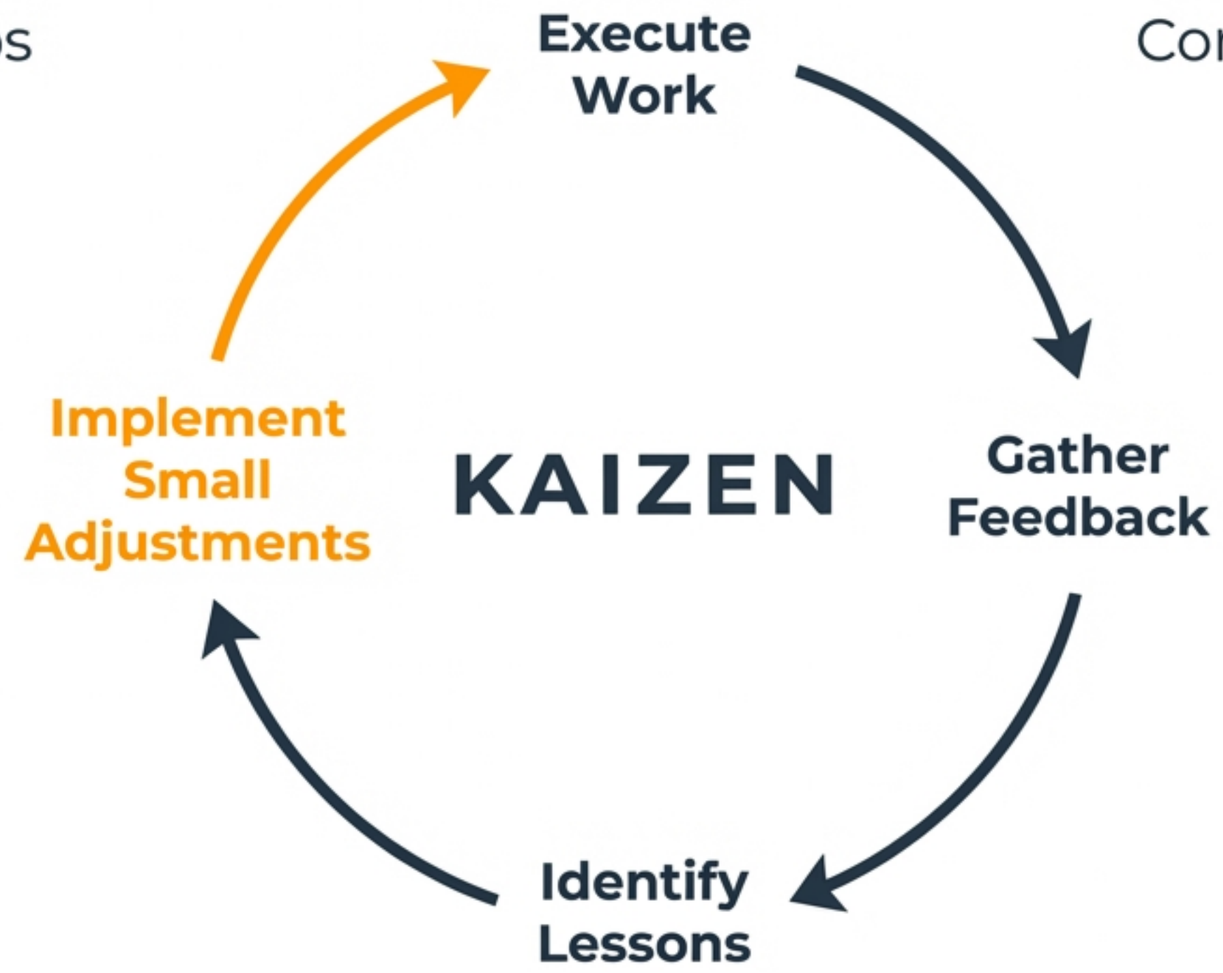
Continuous Improvement (Kaizen)

Feedback loops

Continuous learning



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Real-World Scenario (Defect Issue)

- Rework required
- Cost increased

Situation: Severe Defects Found Late in Project

Increase Inspection ✗

Treats symptom only, high ongoing cost.

Fix Root Cause & Improve Process

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The Exam Architect Workspace

Common Exam Traps

Rely on inspection
Adding more testing → STOP

Focus only on fixing
Fixing the defect only → STOP

Blaming the team → STOP

No improvement loop
Skipping process updates STOP

PMI Mindset: Prevention & Learning

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The Enterprise Mindset

Production systems reliability: Continuous Optimization

Inspection → Monitoring Alerts

Tells you something broke. Damage is done.

Prevention → Fixing Architecture

Engineers the system so failure cannot occur.

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Key Takeaways

Define
Quality = requirements

Design
Prevention > inspection

Invest
Cost of quality matters

Evolve
Improve continuously

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